



ANIMUS TECHNOLOGIES

ANIMUS CORE

Deterministic Telemetry Bridge & Zero-Allocation C++ Middleware for
Mission-Critical Avionics & Low-Latency Execution Pipelines.

<15ns

Lock-free SPSC event logging

ZERO

Dynamic heap allocations

INVARIANT TSC

Hardware cycle timestamping

ZERO-COPY

Python research ingest

AEROSPACE FLIGHT SOFTWARE (GNC • C&DH • HIL) – QUANTITATIVE HFT SYSTEMS

SYSTEMS LEADERSHIP



Alakshendra Roy

Founder & Systems Architect,
Animus Technologies

- Dedicated focus on low-latency bare-metal systems — cacheline topology, lock-free memory models, and deterministic real-time communication bridges.
- Founded Animus to eliminate the friction between high-volume telemetry ingestion and hard-real-time scheduler deadlines in flight avionics and low-latency financial infrastructure.

CORE ENGINEERING TENET

Zero heap allocation on the hot path. Absolute elimination of false sharing. Provable runtime determinism under peak burst load.

The Mechanical Friction in Legacy Telemetry Frameworks

Standard logging architectures — spdlog, ROS2 default loggers, POSIX thread queues — inject tail latency and break determinism by construction, not by misconfiguration.

01 Dynamic Memory Fragmentation

Runtime malloc/new triggers kernel heap locks, page faults, and allocator pauses directly inside the control loop.

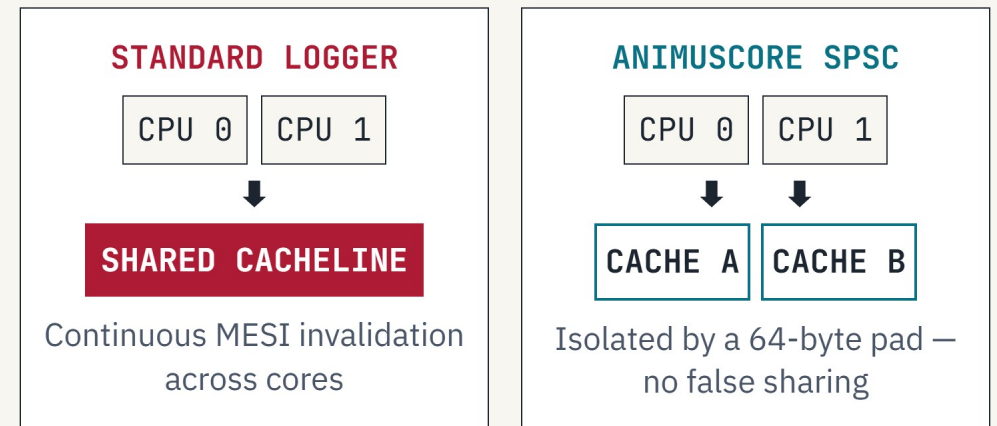
02 Cacheline False Sharing

Producer/consumer pointers and queue metadata sharing a 64-byte boundary force continuous L1/L2 invalidation via MESI bus traffic.

03 Syscall & Clock Jitter

OS timing APIs (std::chrono::system_clock) incur context-switch penalties and scheduler preemption on every call.

CACHELINE TOPOLOGY



AnimusCore Architectural Deep Dive

SINGLE-HEADER C++17

Header-only drop-in implementation, zero external runtime dependencies.

ALIGNAS(64) LOCK-FREE SPSC RING BUFFER

Independent producer/consumer head and tail atomics, isolated across distinct 64-byte boundaries, with explicit acquire/release barriers.

HARDWARE TIMESTAMPING & PIPELINE FENCING

Invariant TSC (constant_tsc / nonstop_tsc) paired with serializing fences (__rdtscp / _mm_lfence) — no OS kernel transition.

DETERMINISTIC OVERFLOW SEMANTICS

Non-blocking overwrite policy: the hot-path producer never spin-waits, never blocks, never allocates.

Runtime isolation. The C++17 single-header engine is reserved for the onboard flight loop / trading critical path. The nanobind zero-copy bridge is decoupled entirely for ground-station ingest, HIL bench inspection, and offline analysis.

Validated Empirical Benchmarks & Tail Analysis

Three logging strategies, one workload: sustained high-rate telemetry emission under burst.

Metric	Standard Logging	Mutex Ring Buffer	AnimusCore SPSC
Median Turnaround	—	—	~100ns
p99 Tail Latency	—	—	—
p99.9 Tail Latency	—	—	<1μs, flat under burst
Max Jitter Spike	—	—	—
Heap Allocations / Sec	—	—	0 (by design)
Max Push Throughput	—	—	16.5M+ pushes/sec

Fill before this goes external: only the bolded AnimusCore figures came from the brief — competitor baselines and the remaining cells are placeholders for real bench output, not estimates. The tail stays flat by construction: a fixed-size, pre-allocated buffer turns burst load into a bounded overwrite, with no lock, allocator, or syscall boundary to stall on.

Methodology: [CPU model] · [compiler + flags] · [kernel / RT-PREEMPT status] · [N trials, warm-up policy]

Minimal Integration Snippets

Hot-path push loop (excerpt)

C++17

```
template <typename T, size_t N>
class SpScRing {
    alignas(64) atomic<uint64_t> head_{0};
    alignas(64) atomic<uint64_t> tail_{0};

    // never blocks, never allocates
    void push(const T& e) noexcept {
        auto pos = head_.load(relaxed);
        buf_[pos & (N-1)] = e;
        head_.store(pos+1, release);
    }
};
```

Ground / HIL zero-copy bridge

PYTHON

```
bridge = animus_core.TelemetryBridge(
    shm_path="/dev/shm/animus_ring0")

# zero-copy view - no heap allocation
events = bridge.as_ndarray(
    dtype=animus_core.TELEMTRY_DTYPE)

latest = events[bridge.head() - 1]
```

CMake

Make

Linux RT-PREEMPT

Custom Aerospace Toolchains

Enterprise Value & Operational ROI

01 • AEROSPACE / NEWSPACE



Aerospace / NewSpace Systems

Eliminates telemetry buffer drops during launch and high-vibration burn phases. GNC and thruster-actuator loops meet hard-real-time deadlines without buffer bloat.

02 • QUANT INFRASTRUCTURE



Quantitative Execution Infrastructure

Fully decouples market-data telemetry and audit logging from the tick-to-trade path, removing scheduler preemption from the execution critical path.

03 • ENGINEERING VELOCITY



Engineering Velocity

Replaces fragile in-house lock-free ring buffer implementations with a verified, hardened telemetry bridge — audited once, reused everywhere.

Evaluation Kit & Sandbox Verification Protocol

01

Evaluation Repo Verification

Compile the evaluation benchmark harness locally.

```
git clone
github.com/alakshendra-
roy/AnimusCore_v1
```

02

Custom Payload Struct Integration

One-week isolated sandbox engagement mapping your telemetry structures into the ring buffer.

03

Bench & Hardware Trace Audit

Direct session with the systems architect to audit latency traces and core-affinity pinning on your HIL testbench or trading infrastructure, under an evaluation agreement.

Evaluation contact – [add scheduling link]